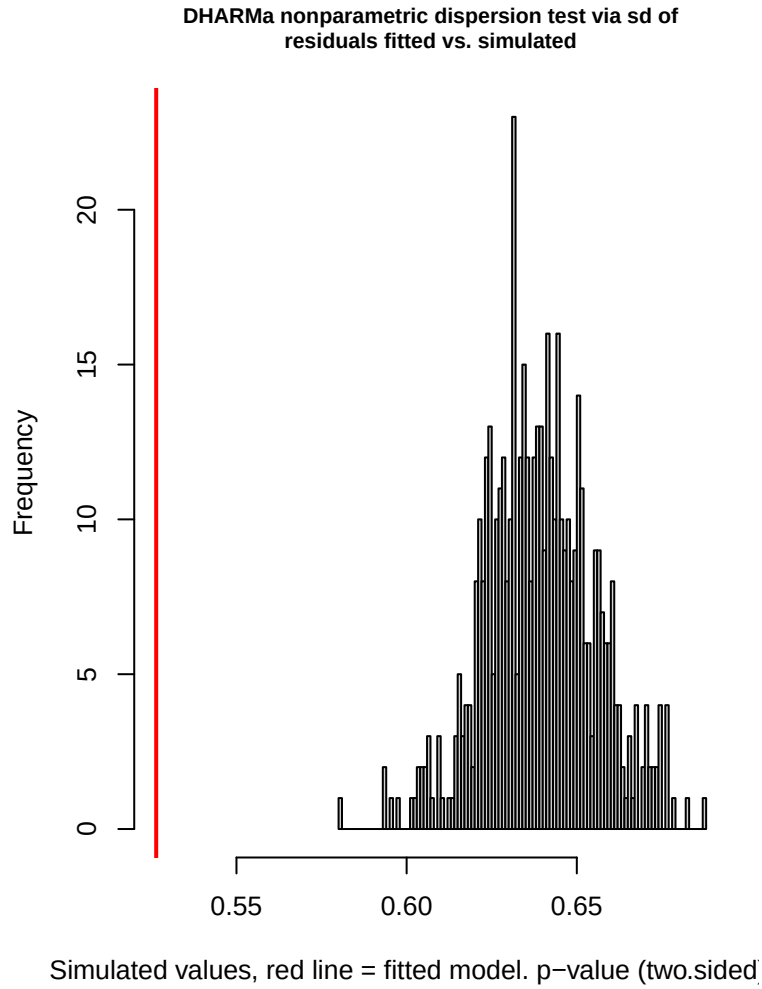
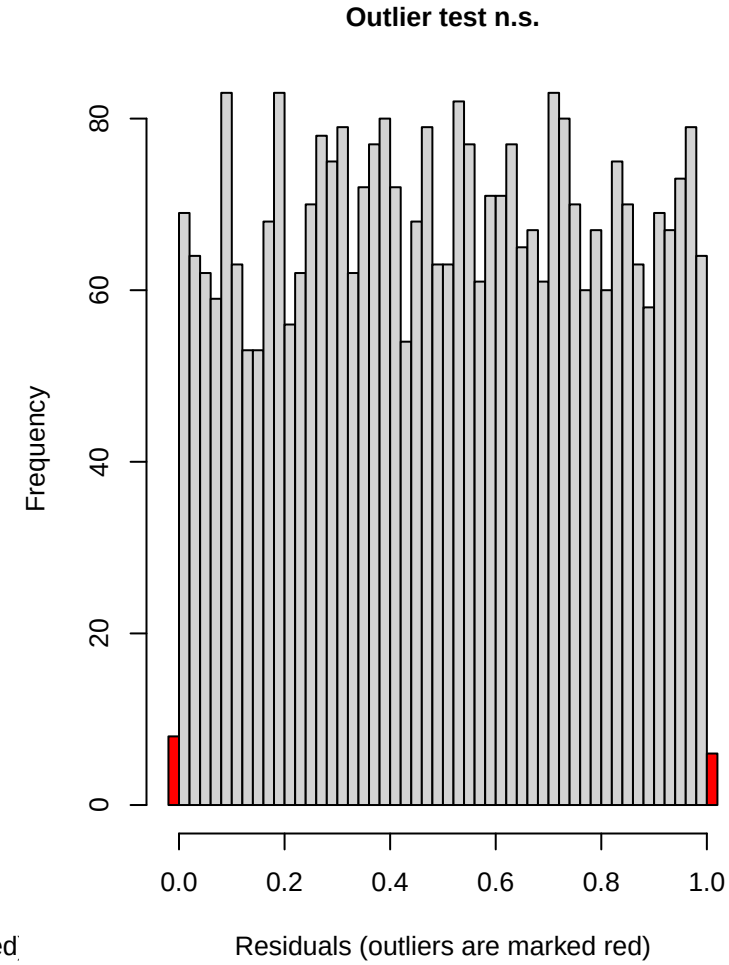
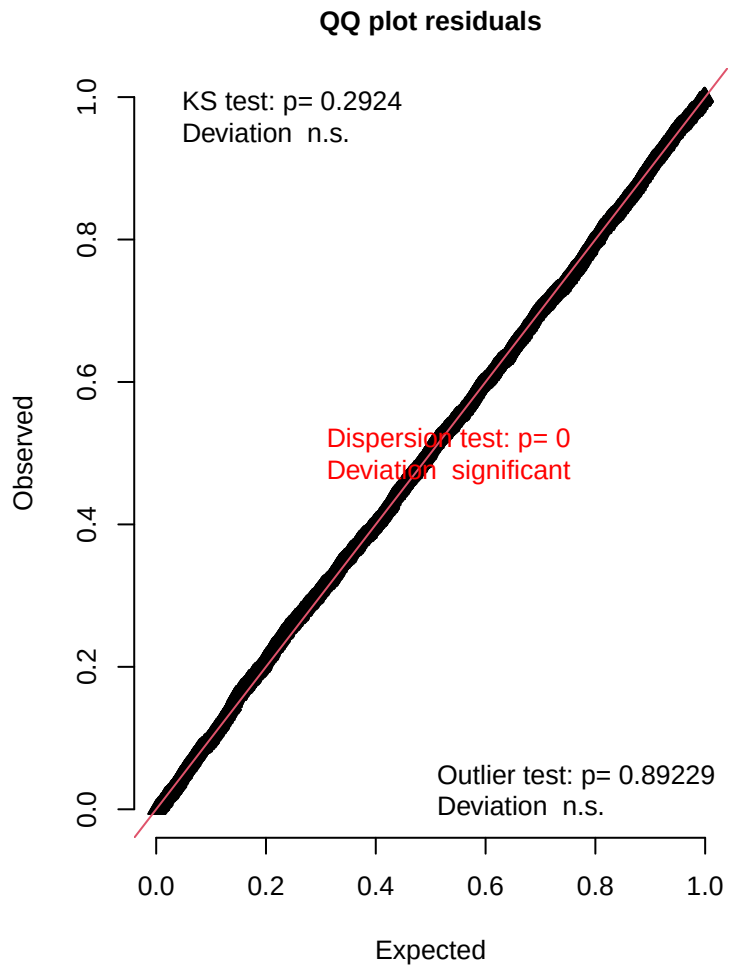
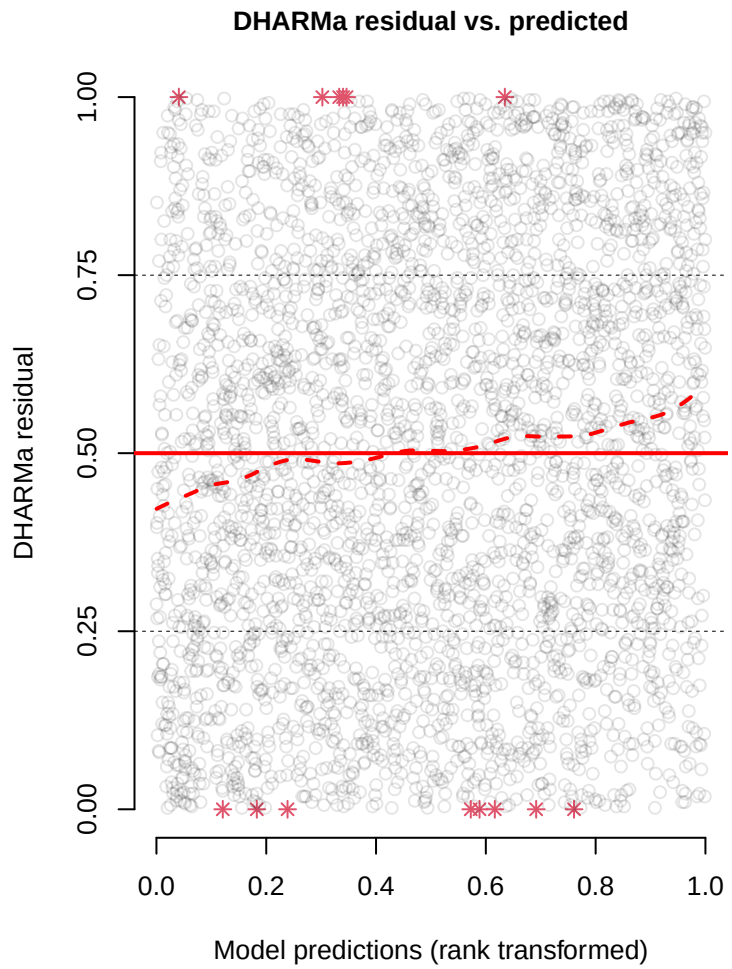
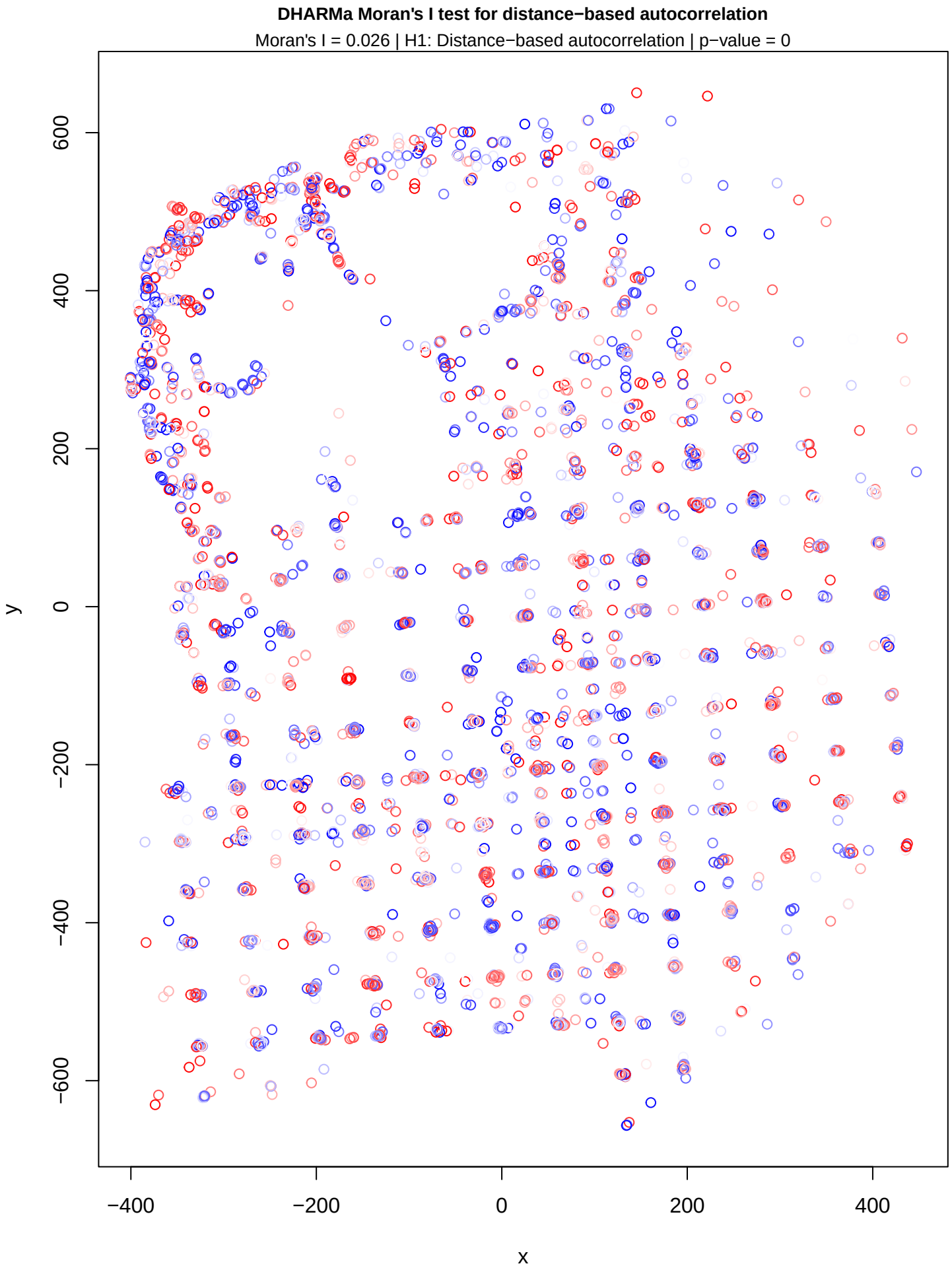
```
--------------------------------------------------------------------
MODEL SELECTION AND VALIDATION REPORT ( general_tric model )
--------------------------------------------------------------------

## OBSERVATIONS:3451

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( general_tric model )

Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

Conditional model:
              coef.est coef.se
(Intercept)       0.06    0.57
sbt_mean          0.28    0.06
sbt_sd           −0.04    0.03
sst_sd           −0.04    0.03
sic_mean          0.03    0.05
log_chla_mean     0.01    0.02
depth             0.03    0.03

Dispersion parameter: 0.76
Spatiotemporal AR1 correlation (rho): 0.97
Matérn range: 517.67
Spatiotemporal marginal AR1 SD: 1.21
REML criterion at convergence: 4305.762

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                  df      AIC   delta_AIC
spatiotemporal_random_fields_ar1                  11 8633.523   3.0218444
spatiotemporal_random_fields_ar1_with_anisotropy 13 8636.545 391.2670753
spatiotemporal_random_fields_iid                  10 9027.812   0.7902226
spatiotemporal_random_fields_iid_with_anisotropy 12 9028.602 142.2512808
spatial_random_fields                             10 9170.854   3.8637225
spatial_random_fields_with_anisotropy             12 9174.717 594.2393356
error_only                                         8 9768.957          NA
```

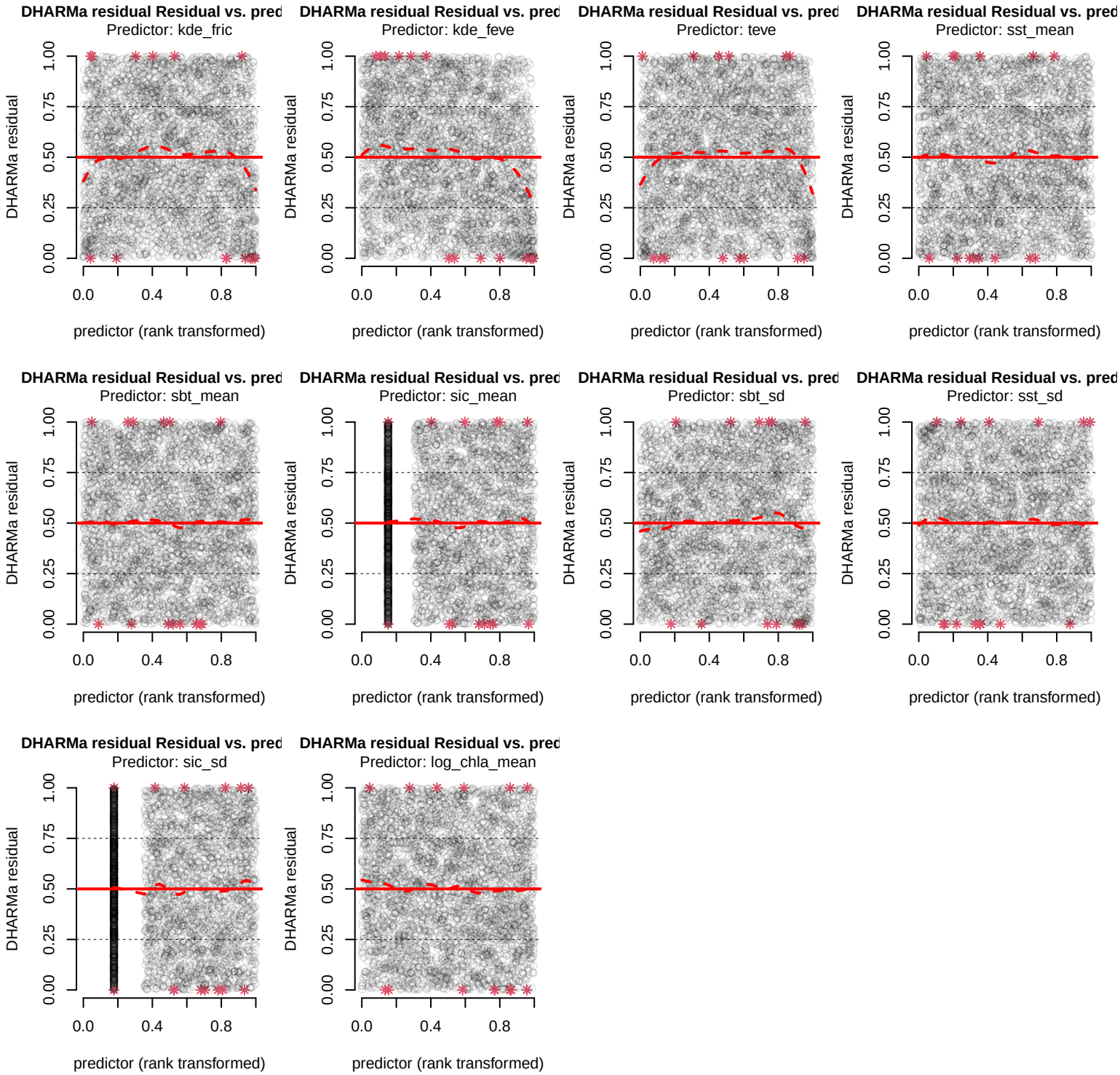

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
20
0.55
0.60
0.65
Simulated values, red line = fitted model. p−value (two.sided


Outlier test n.s.
Frequency
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)


QQ plot residuals
KS test: p= 0.2924
Deviation n.s.
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0.89229
Deviation n.s.
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected


DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.026 | H1: Distance−based autocorrelation | p−value = 0
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_fric
Predictor: kde_feve
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8